THE UNIVERSITY OF HONG KONG

MATH1851 Calculus and Ordinary Differential Equations

Test 1 A (Oct 7, 2022)

Name: _____ **Uni. No.**: _____

Signature: _____

Answer ALL FIVE questions *Time allowed: 45 minutes*

Warning : Students engaged in cheating may be referred to *University Disciplinary Committee.*

Instructions:

- Do **not** tear off any page, and do **not** use your own paper (for rough work).

  No calculator is allowed.

- **Questions 1-3 are multiple choices**:

  Tick *exactly one box for the best answer* in each part.

  "None of them" means "None of the other given options is an answer".

- **Questions 4-5 are short questions**:

  In Parts (a) & (b), give *the answer only.*

  For Part (c), work out according to the instruction in the question.

Use the space below for rough work

1. (a) If $a > 0$ is a constant, then $\lim_{x\to\infty} \dfrac{(2x - \sin ax)(3-x)}{x^2+2x-3} =$

☐ ∞ ☐ -3 ☑ -2 ☐ $a-2$ ☐ 2 ☐ 3 ☐ None of them

(b) Given $\lim_{x\to 0}\left(\dfrac{xf(x)-2}{x-2}\right) = 3$, find $\lim_{x\to 0} f(x)\cdot \sin\dfrac{x}{4}$.

☐ -4 ☐ -2 ☑ -1 ☐ 0 ☐ 1 ☐ 2 ☐ None of them

(c) For what value(s) of b does the limit $\lim_{x\to-\infty}(\sqrt{4x^2+b}+bx)$ exist?

☐ -4 ☐ -2 ☐ 0 ☑ 2 ☐ 4 ☐ 2 or -2 ☐ None of them

(d) Given

c	-1	0	1
$\lim_{x\to c-} g(x)$	-1	2	1
$\lim_{x\to c+} g(x)$	2	-3	-1

. Evaluate $\lim_{x\to 0+} g(\cos x)\cdot g(x)$

☑ -3 ☐ -2 ☐ -1 ☐ 1 ☐ 2 ☐ 3 ☐ None of them

(e) Given

x	-1	0	1
$F(x)$	0	1	-1
$F'(x)$	1	2	3

. Evaluate $G'(0)$ if $G(x) = F(F(x))$.

$G'(0) = F'(F(0))\,F'(0) = F'(1)\cdot 2 = 6$

☐ -3 ☐ -2 ☐ -1 ☐ 1 ☐ 2 ☐ 3 ☑ None of them

2. In Items 2(a)-(c) below, each statement will carry the number of points in the front bracket if the statement is correct, and carry **0 point** if the statement is false. E.g.

(**1 pt**) If $x^2 = 1$, then $x = 1$.

(**2 pts**) If $x = 1$, then $x^2 = 1$.

The first statement carries 0 pt (rather than 1 pt since it is false) while the second statement carries 2 pts (since it is true). Hence we check the box as follows:

The total number of pts is ☐ 0 ☐ 1 ☑ 2 ☐ 3

Use the space below for rough work

2. (a) **(1 pt)** If h has the inverse function $h^{-1}(x)$ and $h(1) = 2, h(2) = 1, h'(1) = 3, h'(2) = 5$,

then $\frac{d}{dx}h^{-1}(x)\Big|_{x=2} = \frac{1}{5}$ $\qquad \frac{1}{h'(h^{-1}(2))} = \frac{1}{h'(1)} = \frac{1}{3}$

(2 pts) If $\lim_{x\to 0+} u'(x) = \lim_{x\to 0-} u'(x) = 2$, then the tangent to $y = u(x)$ at $x = 0$ has slope 2.

The total number of pts is ☑ 0 ☐ 1 ☐ 2 ☐ 3

(b) Use l'hopital's rule, one may compute the limits:

(1 pt) $\lim_{x\to 0} \frac{\sin 2x}{x + \sin x} = \lim_{x\to 0} \frac{2\cos 2x}{1 + \cos x} \cancel{=} \lim_{x\to 0} \frac{-4\sin 2x}{-\sin x} = \lim_{x\to 0} \frac{-8\cos 2x}{-\cos x} = 8$

(2 pts) $\lim_{x\to\infty} (2 + e^x)^{\frac{1}{x}} = 1$

The total number of pts is ☑ 0 ☐ 1 ☐ 2 ☐ 3

(c) Let the function f be continuous and differentiable on $(-\infty, \infty)$.

(1 pt) If $f(0) = -1$ and $y = 1 - x$ is an asymptote to the graph of $y = f(x)$,

then $f(x) = 0$ must have a root in $(-\infty, 0)$.

(2 pts) If the antiderivative F of f has 4 roots (i.e. $F(x) = 0$ has 4 solutions

in $(-\infty, \infty)$), then $f(x) = 0$ will have at least 3 roots.

The total number of pts is ☐ 0 ☐ 1 ☑ 2 ☐ 3

3. Let $f(x) = \begin{cases} \dfrac{(x-4)(x+a)}{|x-4|}, & x < 4, \\ 3a + b|4 - x|, & x \geq 4 \end{cases}$ where a, b are constants and $b \neq 0$.

(a) Evaluate a if f is a continuous function.

☐ -2 ☑ -1 ☐ 0 ☐ 1 ☐ 2 ☐ $b+1$ ☐ None of them

(b) Evaluate the derivative $f'(x)$ for $x < 4$.

☐ $-x$ ☑ -1 ☐ 0 ☐ 1 ☐ x ☐ Not exist ☐ None of them

(c) For what nonzero value(s) of b is f differentiable?

☐ $-2a$ ☑ -1 ☐ 0 ☐ 1 ☐ $2a$ ☐ Not exist ☐ None of them

Use the space below for rough work

4. Consider the curve $\mathcal{C}$ whose equation is $ye^{2x} + x = \sqrt{y}$, and let Q be the y-intercept (i.e. the intersection of $\mathcal{C}$ and the y-axis) which is *not* the origin.

(a) Evaluate the slope of the tangent line at the point Q.

Ans. 6

(b) Suppose f is a function defined on $(-0.1, 0.1)$ such that $(0, f(0))$ is the coordinate of Q and $(x, f(x))$ lies on the curve $\mathcal{C}$ for $x \in (-0.1, 0.1)$. Evaluate the second derivative $f''(0)$.

Ans. 22

(2 marks)

(c) Does there exist any point P on $\mathcal{C}$ at which the tangent is horizontal? Justify your answer with explanation/proof and, if exists, find the coordinate(s) of P.

Ans.

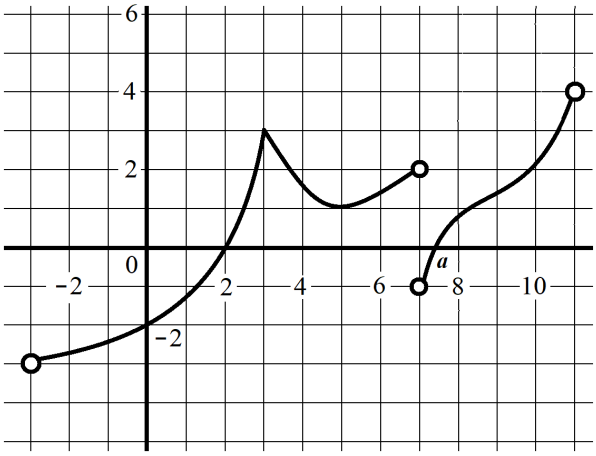
From $ye^{2x} + x = \sqrt{y}$, we get $y'e^{2x} + 2ye^{2x} + 1 = \frac{1}{2\sqrt{y}} y'$.

If $y' = 0$, then $2ye^{2x} + 1 = 0 \Rightarrow y = -\frac{1}{2}e^{-2x} < 0$.

However, the equation $ye^{2x} + x = \sqrt{y}$ is well-defined for $y \geq 0$ only. Thus, there is <u>no</u> point on $\mathcal{C}$ at which $y' = 0$

5. Let H be a continuous function defined on $[-3, 11]$. Besides H is differentiable in $(-3, 7)$ and $(7, 11)$, and the graph of $H'(x)$ is shown in the diagram.

Graph of $y = H'(x)$



($a = 7.4$ denotes the x-intercept of the graph in the interval $(7, 8)$.)

(a) Write down the x-coordinate(s) of the critical point(s) of $H(x)$ in $(-3, 11)$.

Ans. 2, 7, a.
$H'=0$ $H'=0$ H' undefined

(b) Write down the x-coordinate(s) of the following points of $H(x)$ in $(-3, 11)$.

- Local maximum point(s)

Ans. 7

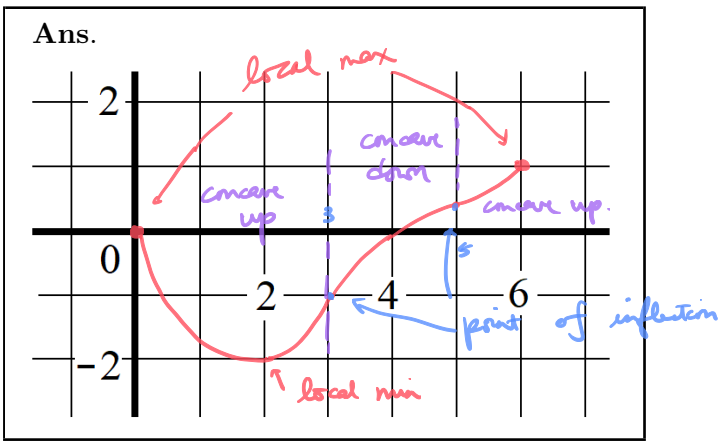
- Point(s) of inflection

Ans. 3,5.

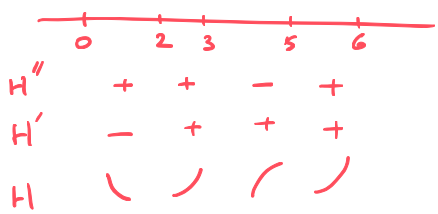
(2 marks)

(c) Given $H(0) = 0$, the minimum and maximum values of H on $[0, 6]$ are -2 and 1 respectively. Sketch the graph of $y = H(x)$ for $0 \le x \le 6$ in the box below. (Indicate in the sketch the concavity, local extrema and inflection point(s).)

Ans.



Rough Work Space



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